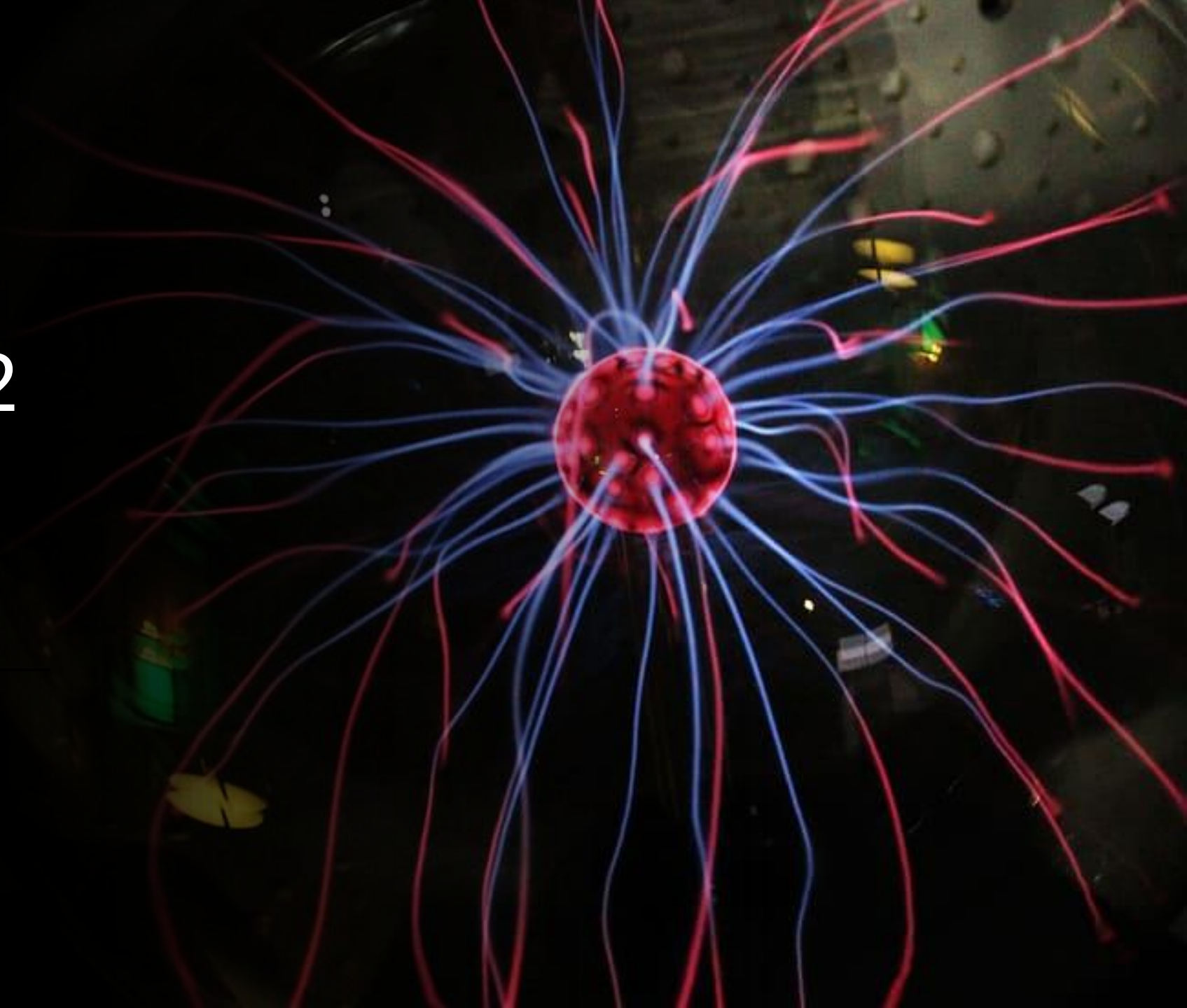




# Mathematics 2 (5EZB0) Lecture 6

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# The gradient

## Gradient

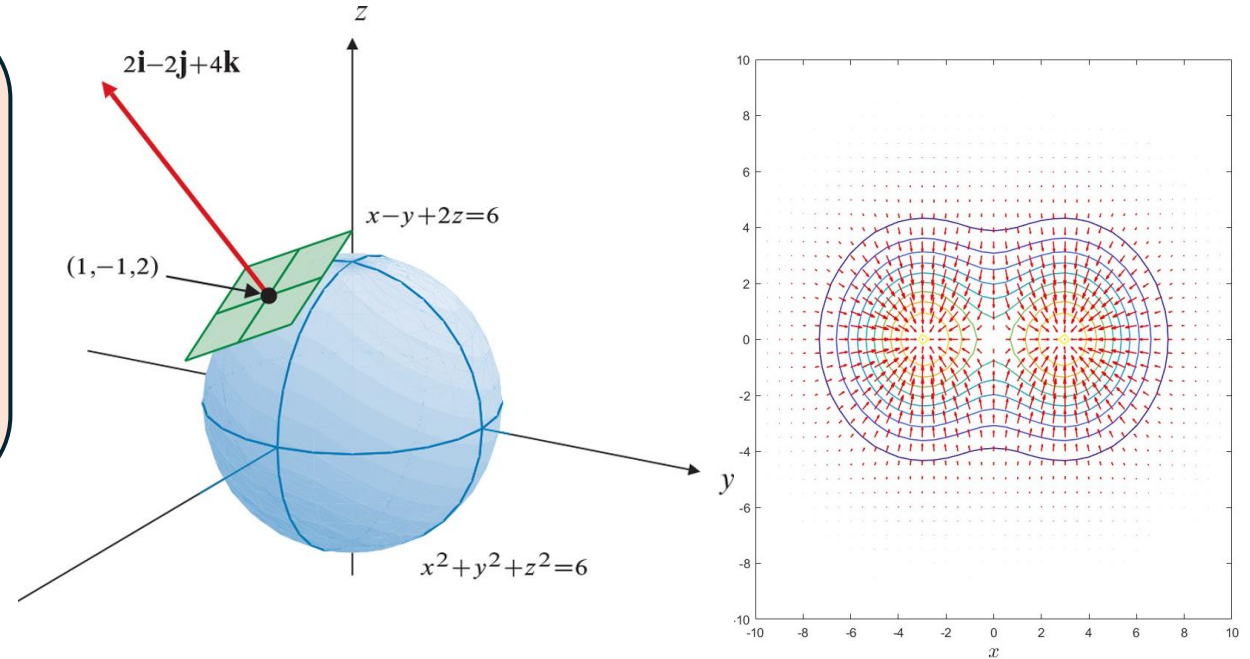
If  $f: \mathcal{D} \rightarrow \mathbb{R}$  is differentiable at a point  $(x_1, \dots, x_d)$

$$\begin{aligned}\nabla f &= \left( \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d} \right) \\ &= \sum_{1 \leq i \leq d} \frac{\partial f}{\partial x_i} \hat{\mathbf{e}}_{x_i}\end{aligned}$$

- The gradient is perpendicular to the level sets
- The  $\nabla f$  is along the direction in which  $f$  increases the fastest, and  $|\nabla f|$  is the maximum increase rate

## Remark

A graph of  $z = f(x, y)$  is the zero level-surface of the function  $w(x, y, z) = f(x, y) - z$



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## Questions 1

What is the curve that passes through  $(1, 1)$  and intersects the level curves of  $f(x, y) = x^3 + y^2$  perpendicularly?

# Important properties of gradient

## Chain rule

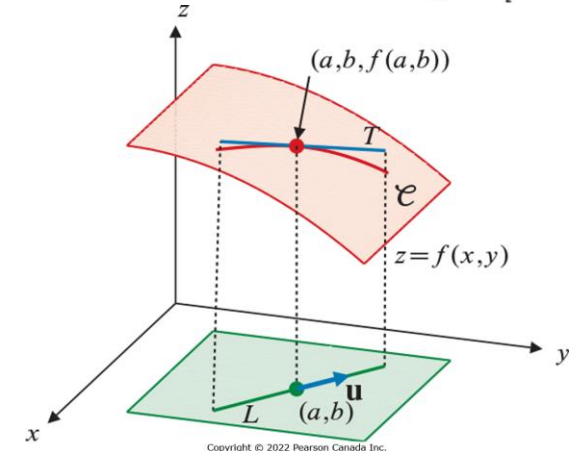
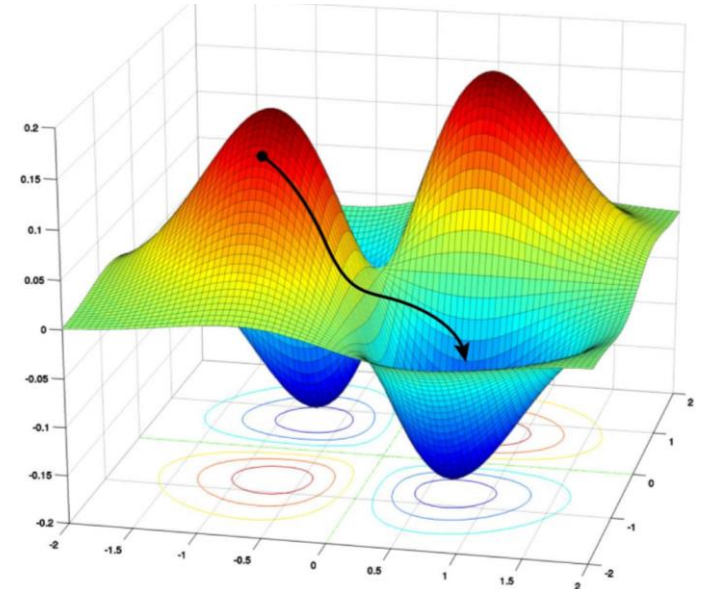
If  $f, x$  are all continuously differentiable, then

$$\frac{\partial}{\partial t} (f(x(t, \cdot))) = \nabla f(x(t, \cdot)) \cdot \frac{\partial x}{\partial t}$$

## Directional derivatives

Directional derivative along a unit vector  $\hat{\mathbf{u}} = (u_1, \dots, u_d)$ .

$$\begin{aligned} D_{\hat{\mathbf{u}}} f(\mathbf{x}) &= \lim_{t \rightarrow 0} \frac{f(x_1 + tu_1, x_2 + tu_2, \dots, x_d + tu_d) - f(x_1, x_2, \dots, x_d)}{t} \\ &= \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t \hat{\mathbf{u}}) - f(\mathbf{x})}{t} = \nabla f(\mathbf{x}) \cdot \hat{\mathbf{u}} \end{aligned}$$



# Vector fields and Jacobian

## Vector fields

For a domain  $\mathcal{D} \subseteq \mathbb{R}^d$ , a vector field  $\mathbf{f}$  is a function of the form

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x}))$$

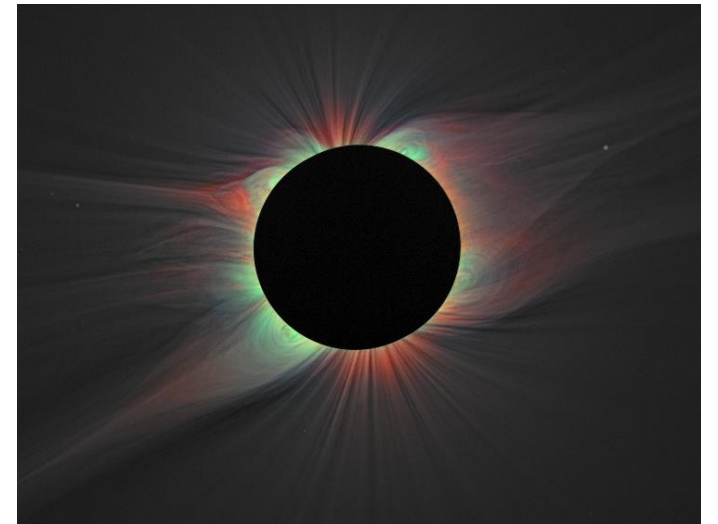
written also as

$$\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^n$$

## The Jacobian

For a domain  $\mathcal{D} \subseteq \mathbb{R}^d$ , consider the vector field  $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^n$  and let it be continuous at  $\mathbf{x} \in \mathcal{D}$ . Then the Jacobian is the  $n \times d$  matrix

$$D\mathbf{f}(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \dots & \frac{\partial f_1}{\partial x_d}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{x}) & \dots & \frac{\partial f_n}{\partial x_d}(\mathbf{x}) \end{pmatrix} \text{ provided all the derivatives exist.}$$



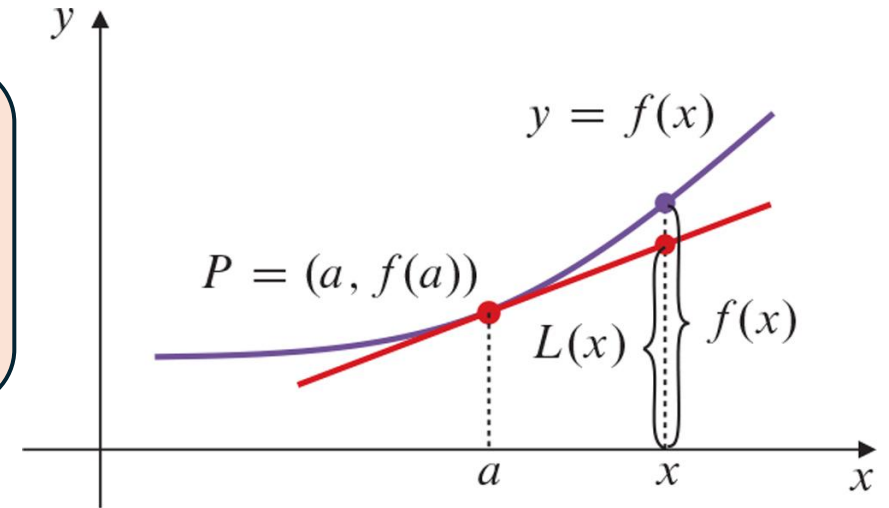
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# Linear approximation

## Mean value theorem

Let  $f: \mathcal{D} \rightarrow \mathbb{R}$  have continuous partial derivatives,  $\mathbf{x}, \mathbf{x} + \mathbf{h} \in \mathcal{D}$  such that the points connecting them lies in  $\mathcal{D}$ . Then, there exists  $\theta \in [0,1]$ ,

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) = \nabla f(\mathbf{x} + \theta \mathbf{h}) \cdot \mathbf{h}$$



## Linear approximation

Let  $|\mathbf{h}| \ll 1$ . Then

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) \approx \mathbf{h} \cdot \nabla f(\mathbf{x})$$

## Questions 2

What is the approximate value of the function

$\mathbf{f}(x, y) = (xe^y + \cos \pi y, x - e^y)$   
at  $(x, y) = (1.02, 0.01)$ ?

# Higher order approximations

## Higher order approximation

Let  $f: \mathcal{D} \rightarrow \mathbb{R}$  have continuous  $(k + 1)^{\text{th}}$  order partial derivatives. Let  $\mathbf{x} \in \mathcal{D}$  and  $\mathbf{h} \in \mathbb{R}^d$  be such that the line connecting  $\mathbf{x}$  and  $\mathbf{x} + \mathbf{h}$  lies in  $\mathcal{D}$ . Then, there exists  $\theta \in [0,1]$ ,

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \mathbf{h} \cdot \nabla f(\mathbf{x}) + \frac{1}{2} (\mathbf{h} \cdot \nabla)^2 f(\mathbf{x}) + \cdots + \frac{1}{k!} (\mathbf{h} \cdot \nabla)^k f(\mathbf{x}) + \frac{1}{(k + 1)!} (\mathbf{h} \cdot \nabla)^{k+1} f(\mathbf{x} + \theta \mathbf{h})$$

## The Hessian

The matrix  $D^2 f := \left( \frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{1 \leq i, j \leq d}$  is called the Hessian matrix

## Questions 3

1. Show that  $D^2 f$  is a symmetric matrix
2. Show that  $D(\nabla f) = D^2 f$
3. Show that  $(\mathbf{h} \cdot \nabla)^2 f = \mathbf{h} \cdot (D^2 f) \mathbf{h}$



# Taylor series

## Taylor series

Let  $f: \mathcal{D} \rightarrow \mathbb{R}$  be a smooth function (all higher order partial derivatives exist and are continuous). If there exists an  $r > 0$ , such that for all  $|\mathbf{h}| < r$ , one has  $\frac{|(\mathbf{h} \cdot \nabla)^k f(\mathbf{x})|}{k!} \rightarrow 0$ , then

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \sum_{k=1}^{\infty} \frac{1}{k!} (\mathbf{h} \cdot \nabla)^k f(\mathbf{x})$$

## Questions 4

Find the second order expansions of the following function around (0,0):

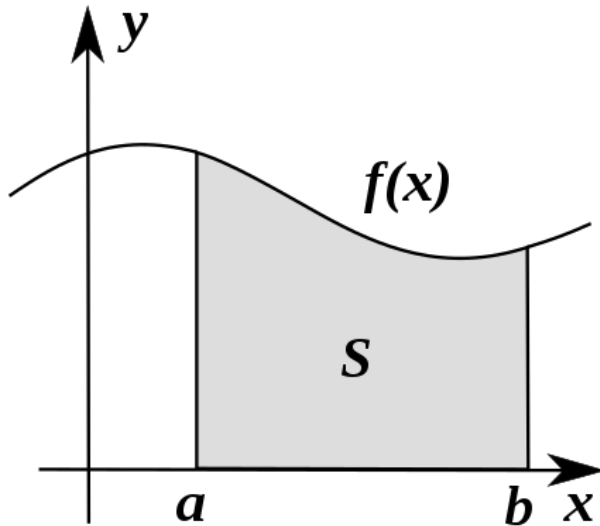
$$f(x, y) = \sqrt{1 + x^2 - y^2}$$

Run **taylor.m** to see the comparison of the function and the approximation

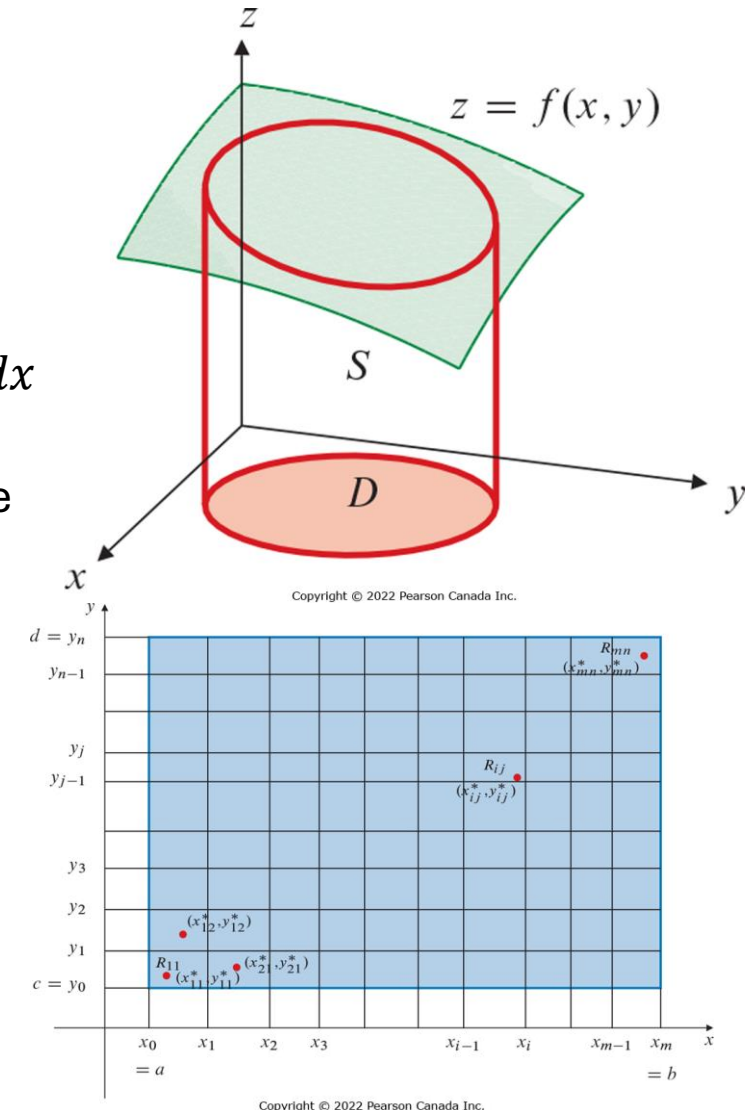
# Double integrals

Finally, we arrive at **Integration!!** We need to give meaning to

$$\iint_{\mathcal{D}} f(x, y) dA$$



- The area under the curve is  $S = \int_a^b f(x) dx$
- The volume under a graph is given by the double integral



Let us first consider the rectangular domain  $\mathcal{D} = [a, b] \times [c, d]$

Consider a partition  $\mathcal{P}$  of  $m \times n$  rectangles  $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ :

$$a = x_0 < x_1 < x_2 < \dots < x_m = b$$

$$c = y_0 < y_1 < y_2 < \dots < y_n = d$$



# Double integrals

- The rectangle  $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$  has area

$$\Delta A_{ij} := (x_i - x_{i-1})(y_j - y_{j-1})$$

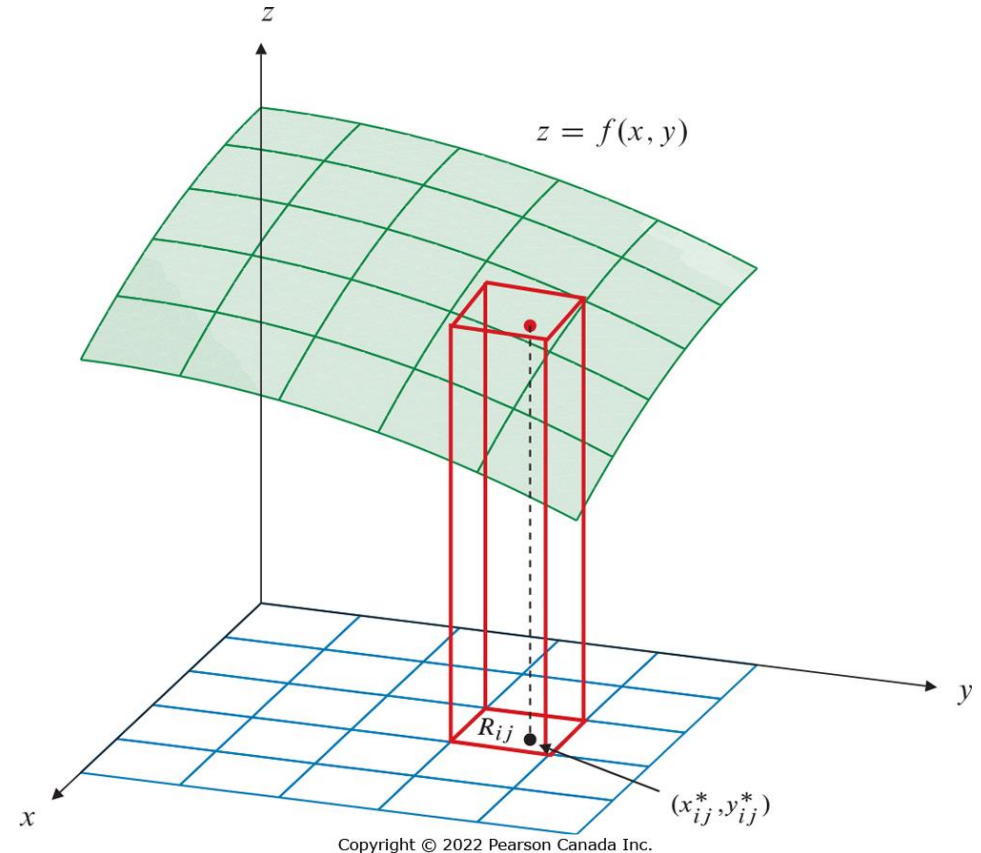
and diagonal  $d_{ij} := \sqrt{(x_i - x_{i-1})^2 + (y_j - y_{j-1})^2}$

and let  $||\mathcal{P}|| := \max_{1 \leq i \leq m} \max_{1 \leq j \leq n} d_{ij}$

- Now consider the *Reimann sum*: for some  $(x_{ij}^*, y_{ij}^*) \in R_{ij}$

$$R(f, \mathcal{P}) = \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$$

- $\int \int_{\mathcal{D}} f(x, y) dA = \lim_{||\mathcal{P}|| \rightarrow 0} R(f, \mathcal{P})$



## Theorem

If  $f: \mathcal{D} \rightarrow \mathbb{R}$  is only discontinuous in a set which has *measure 0*, then the function  $f$  is integrable in  $\mathcal{D}$ , and the integral does not depend on the choice of  $\mathcal{P}$  and the choice of  $(x_{ij}^*, y_{ij}^*)$ .

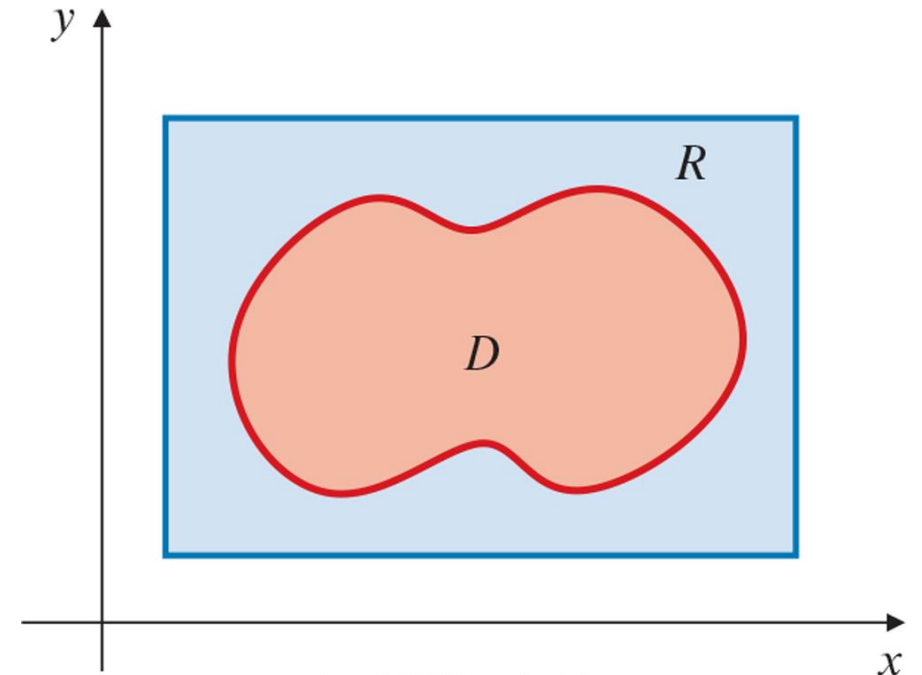
# Riemann integrals

- If  $\mathcal{D}$  is not a rectangle, then consider a rectangle  $R$  surrounding it, and define

$$\tilde{f}(x, y) = f(x, y) \text{ when } (x, y) \in \mathcal{D}, \quad \tilde{f}(x, y) = 0 \text{ otherwise.}$$

Then, if boundary of  $\mathcal{D}$  has measure zero ( $\mathcal{D}$  a *Jordan set*), then

$$\int \int_{\mathcal{D}} f(x, y) dA = \int \int_R \tilde{f}(x, y) dA$$



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## Questions 5

What are the integrals

$$(a) \int \int_{\{x^2+y^2 \leq 4\}} \sqrt{4-x^2-y^2} dA, \quad (b) \int \int_{\{x^2+y^2 \leq 4\}} \sqrt{x^2+y^2} dA$$

Run **DoubleIntegral.py** to find the Riemann sums with different grids on the square  $[-1, 1]^2$

# Properties of the integrals

- If  $\mathcal{D}$  has 0 measure, then  $\int \int_{\mathcal{D}} f(x, y) dA = 0$
- $\int \int_{\mathcal{D}} (\alpha f(x, y) + \beta g(x, y)) dA = \alpha \int \int_{\mathcal{D}} f(x, y) dA + \beta \int \int_{\mathcal{D}} g(x, y) dA$  **(Linearity)**
- $\int \int_{\mathcal{D}} f(x, y) dA \leq \int \int_{\mathcal{D}} g(x, y) dA$  if  $f(x, y) \leq g(x, y)$  in  $\mathcal{D}$  **(Preserving ordering)**
- $|\int \int_{\mathcal{D}} f(x, y) dA| \leq \int \int_{\mathcal{D}} |f(x, y)| dA$  **(Triangle inequality)**
- Let  $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_k$  be disjoint domains. Then **(Additivity)**

$$\int \int_{\cup_{i=1}^k \mathcal{D}_i} f(x, y) dA = \sum_{1 \leq i \leq k} \int \int_{\mathcal{D}_i} f(x, y) dA$$

## Questions 6

Find the integral (b) from Question 5 in the annulus  $\{1 \leq x^2 + y^2 \leq 2\}$